

Astron 98 Final Project: Journey Through The Universe

1) Introduction

The goal of this project was to create a scientifically meaningful and intuitive visualization of the size differences between physical structures in the universe. The universe spans over 60 orders of magnitude, from the Planck length ($\sim 10^{-35}$ m) to the diameter of the observable universe ($\sim 10^{26}$ m). Humans struggle to comprehend these extreme scales, so this project uses Python to animate size comparisons interactively and educationally.

2) Scaling (Mathematics)

To compare objects with sizes differing by dozens of orders of magnitude, direct scaling is impossible. If I drew objects at true scale, almost every circle would disappear or fill the screen. Instead, I used a compressed relative-radius formula:

$$R = R_{base} \left(\frac{S_{other}}{S_{current}} \right)^{0.3}$$

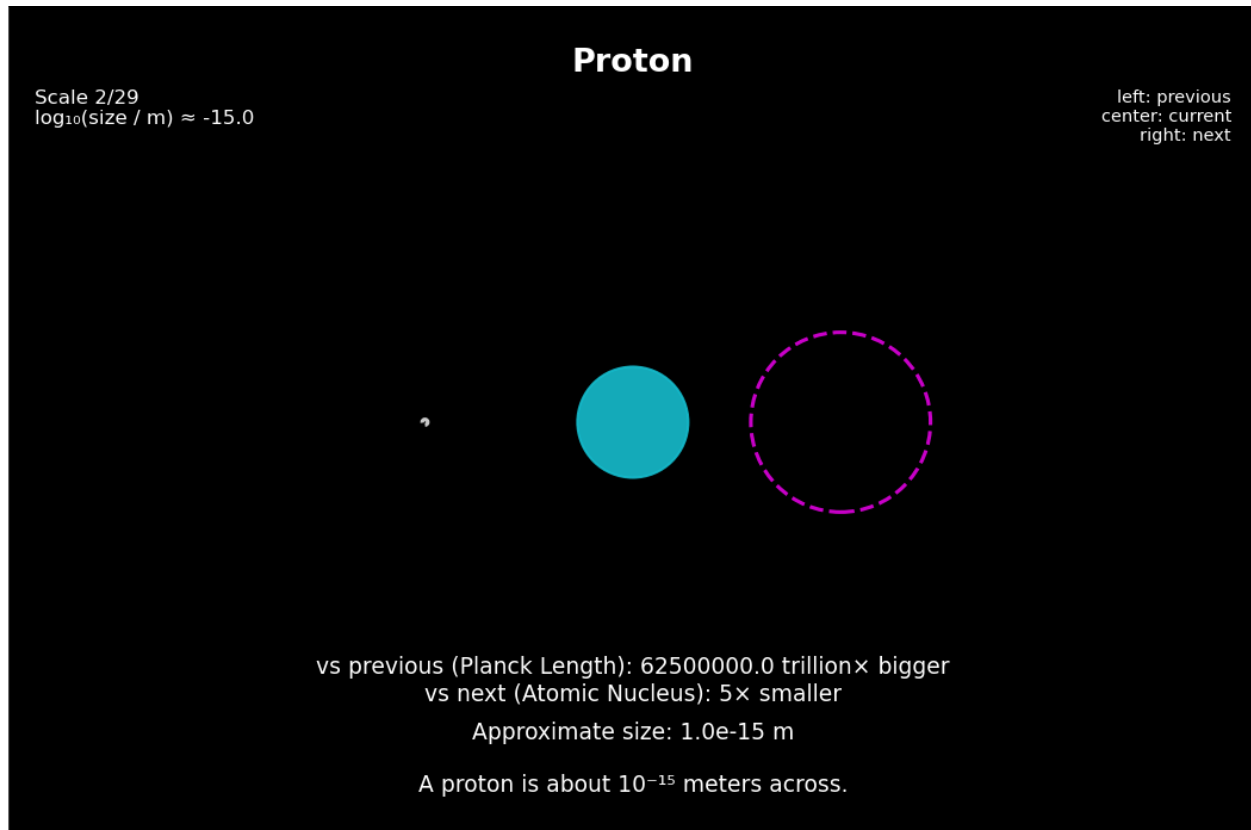
Where R_{base} is the fixed radius of the current object, $S_{current}$ and S_{other} are the physical sizes (in meters). The exponent functions similarly to a logarithmic transform, allowing scale differences of 10^6 – 10^{20} to appear without breaking the visualization.

3) Steps

Libraries Used: numpy, matplotlib, matplotlib.animation, math

The project began by defining a dataset of 29 structures in the universe, each with: A name, physical size in meters, and two “fun facts”. I stored these in a list of dictionaries.

Next, I created the animation. The current object is drawn as a solid cyan circle. The previous scale is drawn as a dashed white circle, and the next scale as a dashed magenta circle. The animation is controlled through keyboard input. To prevent circles from clipping off the screen (a problem that appeared once objects reached galactic scales), I implemented an “auto-zoom” viewport that adapts to the largest circle shown.



Lastly, I added text overlays for the current level $\log_{10}(\text{size})$, comparison ratios, fun facts, and a scale indicator.

4) Animation

The animation updates as the user presses the arrow keys. Each frame rescales the view, updates all circle radii, and refreshes the displayed text

Here is a description of the animation output:

- Objects appear side-by-side for visual comparison.
- Smaller objects shrink proportionally, but remain visible because of our radius compression.
- Larger objects expand and recenter with the adaptive viewport.
- Fun facts rotate with “n” and “p” keys. Ratio descriptions appear below the circles.

Video Link to Animation: <https://youtu.be/v8Xgi8ylPHE>

5) Challenges

I had several technical difficulties. My main difficulty was keeping the objects visible and properly scaled, as some objects differed by up to 10^{20} . Without compressed scaling or auto-zoom, the animation would break. Alongside that, large objects would be so large as to disappear from the screen, missing the point of my project. Another difficulty that I faced was formatting ratios. I wanted my project to be easy to read and understand, so that the average viewer could follow it. I had to write a formatter that turns numbers such as $3.4e+12$ into “3.4 trillion”. Additionally, I ran into issues with the update function. I had to ensure that only the frame index passed through the callback since additional parameters caused errors.

6) Conclusion

This project demonstrated how computational tools can help communicate concepts that are otherwise abstract and difficult to visualize. By using Python to animate the scale of the universe, I created an educational tool that highlights the enormous size differences between structures around us.

7) Work Cited

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